ST115
Lab: Data Visualization

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Lab Today

- This is a guided lab activity that will introduce you to some new functions
  - case_when()
  - scales_fill_manual()
  - Themes in ggplot
- I will help you through most questions
- If we don't get to the last question, Dr. Meyer will cover it in class
- Assignment due Friday night

The Set-Up

- Clone the durham-climate repo from GitHub
- Download the durham-climate.csv from Moodle
- Create a data folder in your data-viz-lab repo
- Put the durham-climate.csv into the data folder you just created
- Read in the data using read_csv()! Name your data durham_climate

```
durham_climate <- read_csv("your/correct/file/path/durham-climate.csv")
```

fct_relevel

Run the following code in your **Console**. What is it telling us?

```
fct_unique(durham_climate$month)
```

("Reorder - Fix the plot" part of lab)

fct_relevel

Run the following code in your **Console**. What is it telling us?

fct_unique(durham_climate$month)

```
Levels: April August December February January July June March May November October September
```

We can see how the levels of the month variable are ordered. This is why the plot is incorrect. We can fix this with fct_relevel()

("Reorder - Fix the plot" part of lab)

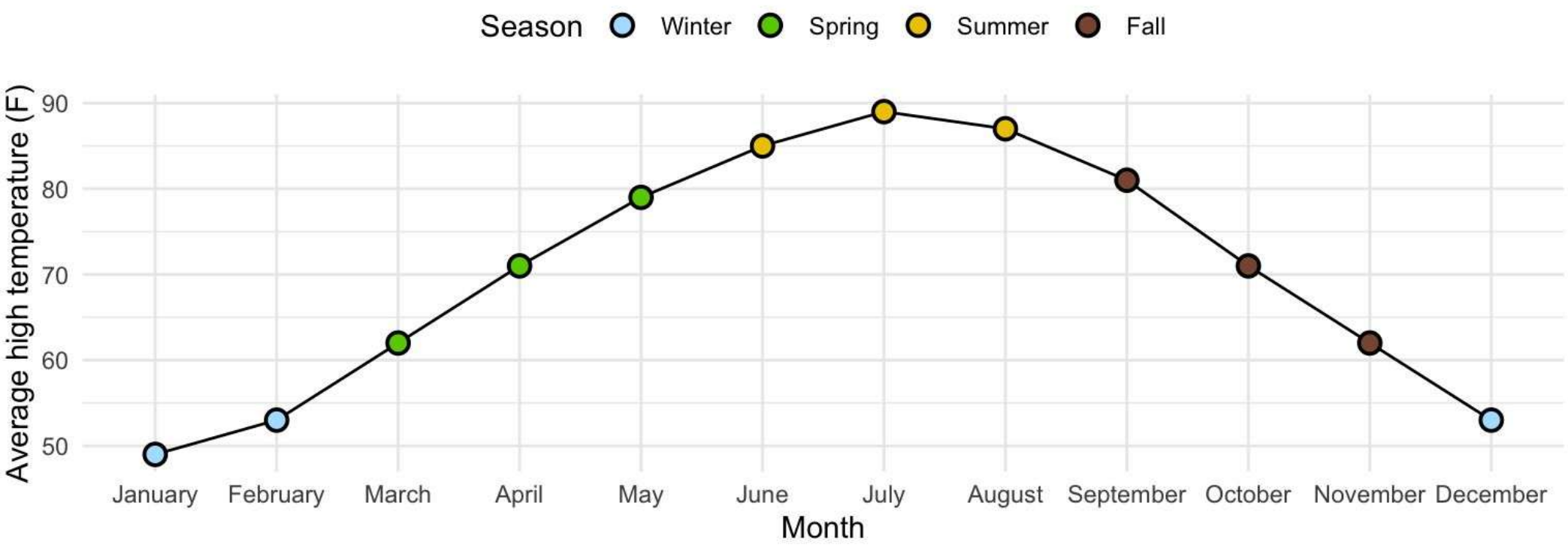
fct_relevel

- fct_relevel()
  - The first argument is the name of the variable you want to relevel
  - The second argument is how you want to relevel that variable
- Questions to ask yourself
  - What dplyr function do we use to create/change columns?
  - How can the pre-defined month.name object help us?

("Reorder - Fix the plot" part of lab)

This is the plot that we want. What information do we need that we don't have in the data set?

("Recode and Reorder" part of lab)



case_when

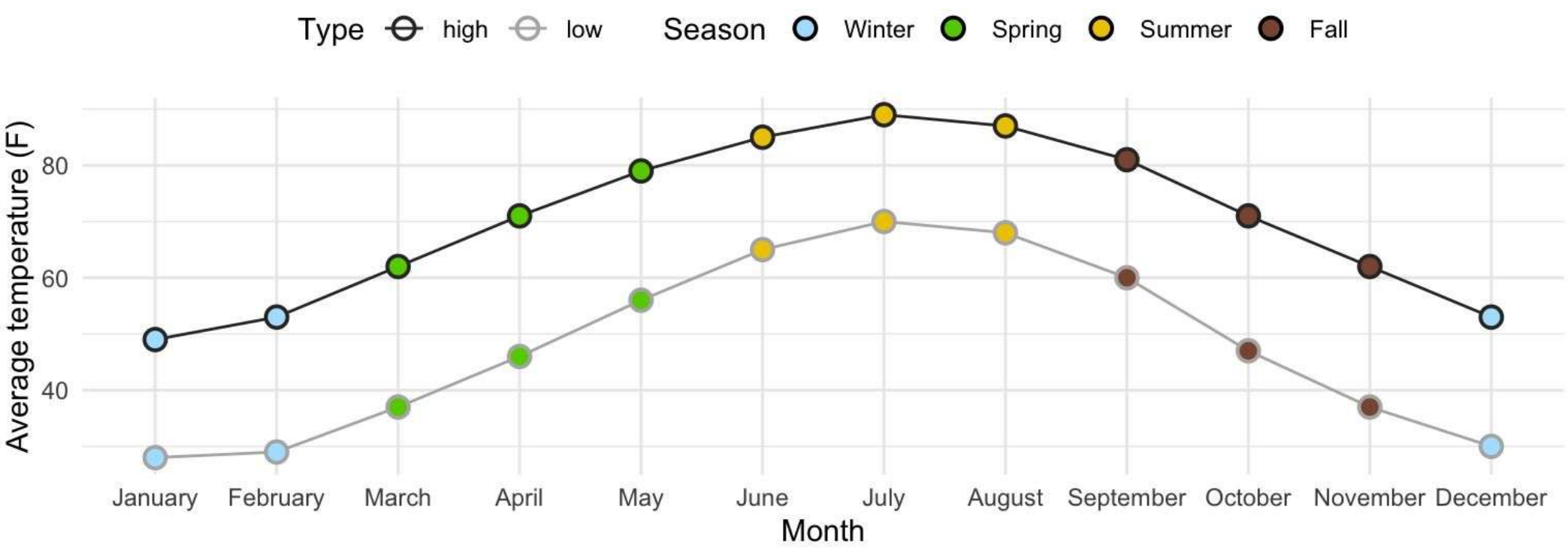
Let's practice reading this code together as a sentence, and think about how this code is helping us update our plot, coloring points based on season.

```
durham_climate <- durham_climate |>
  mutate(
    season = case_when(
      month %in% c("December", "January", "February") ~ "Winter",
      month %in% c("March", "April", "May") ~ "Spring",
      month %in% c("June", "July", "August") ~ "Summer",
      month %in% c("September", "October", "November") ~ "Fall",
    ),
    season = fct_relevel(season, "Winter", "Spring", "Summer", "Fall")
  )
```

("Recode and Reorder" part of lab)

This is the plot that we want. What information do we need that we don't have in the data set?

("Recode and Reorder" part of lab)



Try it on your own first!

("Recode and Reorder" part of lab)

Let's read this code together as a class. Run it. How is this helping us?

```
durham_climate |>
  pivot_longer(
    cols = c(avg_high_f, avg_low_f),
    names_to = "temp_type",
    names_prefix = "avg_",
    values_to = "avg_temp_f"
  )
```

("Pivot" part of lab)